Fundamentals of Fluid Mechanics, 6th Edition Paperback – 4 November 2009

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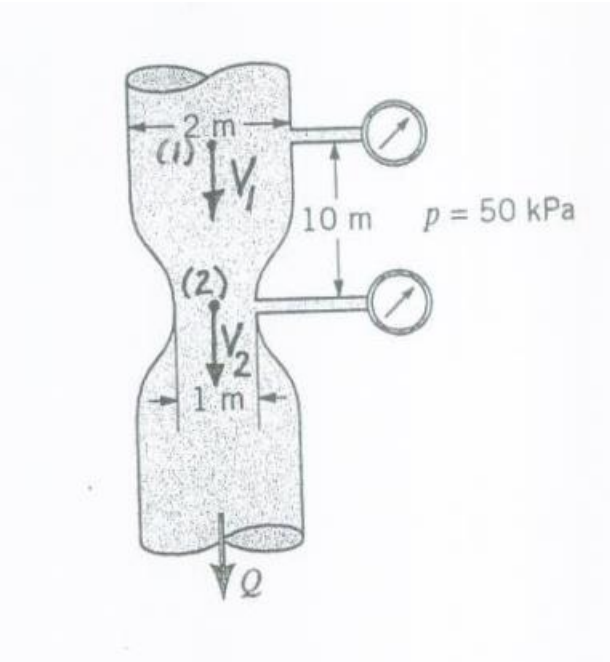
Questions from the book mentioned above. I have mentioned question numbers from the book for most questions. You can find the solutions in the manual. If I did not mention the question number, find the solution in the manual yourselves.

1. Water flows steadily through a variable area horizontal pipe. The velocity is given by $\bar{V} = 10(1+x)\bar{\imath}$, where x is in meters and velocity is in m/s. Coordinate x is zero at the start of the pipe section and positive towards left. Viscous effects are neglected. Determine the pressure at the end of the pipe as a function of x. If the length of the pipe is 10 m, what is the pressure at the end? If the pressure at the starting point is 70 KPa, what is the pressure at the end?
   3.3
2. What pressure gradient along the streamline is required to accelerate the water in a horizontal pipe at a rate of 30 m/s^2?
   3.6
3. Consider a compressible fluid for which the pressure and density are related by $\frac{P}{\rho^n} = C_0$, where C_0 and n are constants. Derive the Bernoulli equation for steady and inviscid flow along the streamline.
   3.9
4. When an airplane is flying 350 kmph at 1600 m altitude in a standard atmosphere, the air velocity at a certain part of the wing is 400 kmph relative to the airplane. What suction pressure develops at this point? What is the pressure at the leading edge (a stagnation point) of the wing?
   3.18
5. A person thrusts his hand into the water while travelling 3 m/s in a motor boat. What is the maximum pressure on his hand?
   3.24
6. Water flows through a hole in the bottom of a large, open tank with a speed of 10 m/s. Determine the depth of the water in the tank. Viscous effects are negligible and the diameter of the hole is very small compared to tank diameter.
7. Water flows from a garden hose with a velocity of 15 m/s. What is the maximum height that it can reach above the nozzle? The flow is vertically upwards.
8. Water flows steadily in the vertical tube area pipe shown in fig. Determine the flow rate if the pressure in each of the gages reads 50 kPa.



3.47

9. 3.56
The circular stream of water from a tap is observed to observed to taper from a diameter of 20 mm to 10 mm in a distance of 50 cm. Determine the flow rate.

3.59 A smooth plastic, 10-m-long garden hose with an inside diameter of 20 mm is used to drain a wading pool as is shown in Fig. P3.59. If viscous effects are neglected, what is the flowrate from the pool?

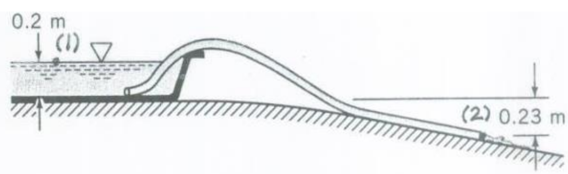


FIGURE P3.59

$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2$$ where $p_1 = p_2 = 0$, $z_1 = 0.2m$

$z_2 = -0.23m$, and $V_1 = 0$

Thus,

$$V_2 = \sqrt{2g(z_1 - z_2)} = \left(2(9.81\,\tfrac{m}{s^2})(0.2\,m - (-0.23m))\right)^{\frac{1}{2}}$$

$$= 2.90\,\frac{m}{s}$$

or

$$Q = A_2 V_2 = \frac{\pi}{4}(0.020m)^2(2.90\,\tfrac{m}{s}) = 9.11 \times 10^{-4}\,\frac{m^3}{s}$$

3.60 Water exits a pipe as a free jet and flows to a height h above the exit plane as shown in Fig. P3.60. The flow is steady, incompressible, and frictionless. (a) Determine the height h. (b) Determine the velocity and pressure at section (1).

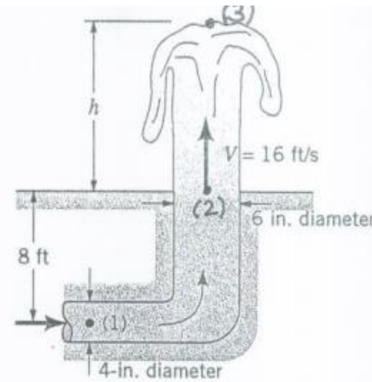


FIGURE P3.60

(a) From the Bernoulli eqn.,

$$\frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 = \frac{p_3}{\gamma} + \frac{V_3^2}{2g} + z_3, \text{ where } p_2 = p_3 = 0, \text{ and } V_3 = 0.$$

Thus,

$$\frac{V_2^2}{2g} = z_3 - z_2 = h$$

or

$$h = \frac{V_2^2}{2g} = \frac{(16\ ft/s)^2}{2(32.2\ ft/s)} = \underline{\underline{3.98\ ft}}$$

(b) Also, $A_1 V_1 = A_2 V_2$

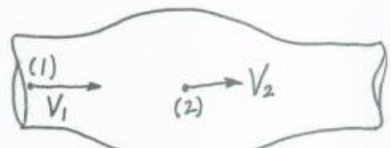
or
$$V_1 = \frac{A_2}{A_1} V_2 = \frac{\frac{\pi}{4}(6\text{ in.})^2}{\frac{\pi}{4}(4\text{ in.})^2}\left(16\,\frac{ft}{s}\right) = \underline{\underline{36.0\,\frac{ft}{s}}}$$

From the Bernoulli equation,
$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2,$$

or since $\gamma = \rho g$,
$$p_1 = p_2 + \frac{1}{2}\rho(V_2^2 - V_1^2) + \gamma(z_2 - z_1) \quad \text{where } p_2 = 0$$

Thus,
$$p_1 = \frac{1}{2}\left(1.94\,\frac{slugs}{ft^3}\right)\left[\left(16\,\frac{ft}{s}\right)^2 - \left(36.0\,\frac{ft}{s}\right)^2\right] + 62.4\,\frac{lb}{ft^3}(8\,ft)$$
$$= -1009\left(\frac{slugs \cdot ft}{s^2}\right)/ft^2 + 499\,\frac{lb}{ft^2}$$
$$= \underline{\underline{-510\,\frac{lb}{ft^2}}}$$

3.62 Blood ($SG = 1$) flows with a velocity of 0.5 m/s in an artery. It then enters an aneurysm in the artery (i.e., an area of weakened and stretched artery walls that cause a ballooning of the vessel) whose cross-sectional area is 1.8 times that of the artery. Determine the pressure difference between the blood in the aneurysm and that in the artery. Assume the flow is steady and inviscid.



$A_2 = 1.8 A_1$

From the Bernoulli equation,

$$p_1 + \tfrac{1}{2}\rho V_1^2 + \gamma z_1 = p_2 + \tfrac{1}{2}\rho V_2^2 + \gamma z_2$$

where $z_1 = z_2$ and $V_1 = 0.5 \frac{m}{s}$

Thus,

$$p_2 - p_1 = \tfrac{1}{2}\rho\left(V_1^2 - V_2^2\right)$$

However,

$$\rho = \rho_{H_2O}\, SG_{blood} = \rho_{H_2O}(1) = 999 \frac{kg}{m^3}$$

and

$$V_1 A_1 = V_2 A_2 \text{ or}$$

$$V_2 = \frac{A_1}{A_2} V_1 = \left(\frac{1}{1.8}\right) V_1$$

Thus, Eq (1) becomes

$$p_2 - p_1 = \tfrac{1}{2}\left(999 \frac{kg}{m^3}\right)\left[\left(0.5 \frac{m}{s}\right)^2 - \left(\frac{1}{1.8}\right)^2\left(0.5 \frac{m}{s}\right)^2\right]$$

$$= 86.3\left(\frac{kg \cdot m}{s^2}\right)/m^2 = 86.3 \frac{N}{m^2} = \underline{\underline{86.3\ Pa}}$$